We applied this transition 10% of the times, and there we go, the average crossing number went down to 19.17: meaning that the algorithm was almost always finding the best solution. Speaking of, figure 17.14 compares two solutions for the embedding of K_8: the one on the left was found by the random sampling algorithm provided in chapter 16, and the other one is the result of the simulated annealing-based method.

It goes without saying, there could be a lot more work to do to improve the algorithm by fine-tuning the parameters and perhaps coming up with better operators to tweak the solutions.

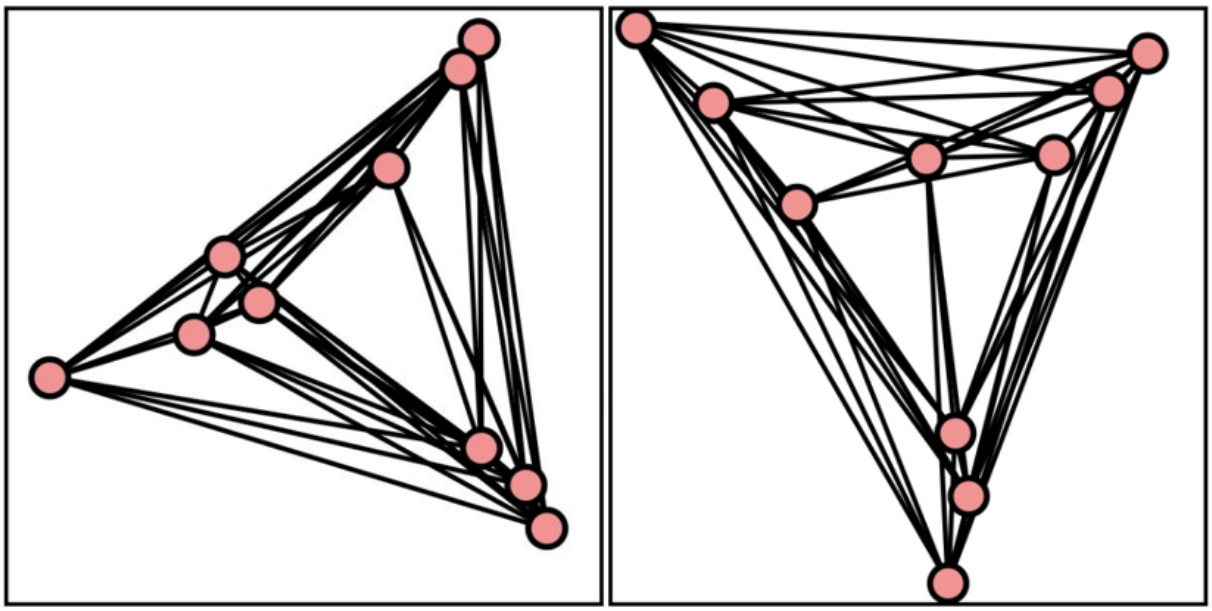


Figure 17.15 Two embeddings of complete graph K_{10}: simulated annealing (right) produced an optimal embedding with 62 intersections, while the one outputted by random sampling has 81 intersections.